

MAC F244: Stochastic Calculus and its Applications in Finance - Course Project

Delta Hedging under Stochastic Volatility

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Abstract

We present a unified study of reinforcement learning for dynamic delta hedging, integrating deep learning-based volatility forecasting and classical stochastic calculus theory. Three components are developed and connected end-to-end.

Component 1 trains a Proximal Policy Optimization (PPO) agent to hedge a European call option under geometric Brownian motion (GBM), benchmarked against the analytically optimal Black-Scholes-Merton (BSM) delta. The standard agent achieves a BSM-delta correlation of 0.947 and mean absolute hedging error (MAE) of 0.830; six theory-motivated extensions are evaluated, with domain randomisation ($\sigma \sim U(0.10, 0.40)$) achieving the best RL MAE of **0.723** and highest BSM correlation of **0.963**.

Component 2 forecasts 21-day forward realised volatility on Nifty 50 using a two-layer stacked LSTM, achieving out-of-sample MAE 0.0114 and RMSE 0.0203, outperforming GARCH(1,1) by 41% on MAE and 20% on RMSE. A naive rv21 persistence baseline (MAE 0.0043) outperforms all models at the 21-day horizon due to volatility clustering, and serves as the correct explanation for the pricing horse-race results.

Integration connects both components: an LSTM-informed RL agent trained in a varying- σ environment achieves **MAE = 0.407** versus 1.435 for a constant- σ baseline when evaluated in their shared natural habitat (72% reduction, $p < 0.001$ by paired bootstrap). On 51 non-overlapping windows of real Nifty paths spanning 2020–2024, BSM daily hedging with rv21 (MAE 105.1) outperforms RL (MAE 137.5, $p = 0.013$); replacing rv21 by the LSTM forecast improves the analytic hedge to MAE 96.1, and a mixed partial-rebalancing rule around the BSM-LSTM target achieves the best real-path MAE of 88.3. All strategies deteriorate sharply above transaction cost $\kappa = 0.003$.

1 Introduction

1.1 Motivation and Context

The Black-Scholes-Merton (BSM) model [2, 11] gives a closed-form price for European options and an exact hedging rule under idealised assumptions: frictionless continuous trading and constant volatility. Two of those assumptions fail simultaneously in practice:

1. **Transaction costs and discrete rebalancing.** Continuous trading is costless in theory; daily rebalancing with proportional costs $\kappa|\Delta\delta|\cdot S$ creates a genuine optimisation problem whose solution has no known closed form. The RL agent’s task is to discover this solution.
2. **Stochastic volatility.** Equity volatility is random and mean-reverting. Under a stochastic-vol model such as Heston [7], BSM delta computed at a fixed σ is provably suboptimal. An agent that observes the current variance state can, in principle, do better.
3. **Unknown volatility.** Even if BSM were correct, σ is unobservable and must be forecast. Forecast accuracy translates directly to pricing and hedging quality via the Vega Greek.

This project addresses all three layers within a single computational framework, with stochastic calculus as the mathematical spine throughout.

1.2 Structure and Contributions

The project is organised into five parts, all sharing the BSM pricing engine and Greeks (Δ , Γ , $\mathcal{V}$):

- Part I** PPO agent for delta hedging under GBM and Heston, with BSM and bootstrap baselines.
- Part II** Six novel RL experiments varying reward function (MSE vs. Itô-decomposed), environment (GBM vs. Heston), observation (with/without gamma), and training distribution (fixed vs. domain-randomised σ).
- Part III** Stacked LSTM for Nifty 50 forward volatility forecasting, benchmarked against GARCH(1,1), a naive rv21 persistence baseline, and India VIX implied volatility.
- Part IV** LSTM–RL integration with a natural-habitat evaluation that corrects a distributional alignment issue present in naive comparison.
- Part V** Extended real-path backtesting: 51 non-overlapping windows, including analytic, RL, and mixed BSM–LSTM execution rules, plus transaction-cost sensitivity and moneyness stress tests.

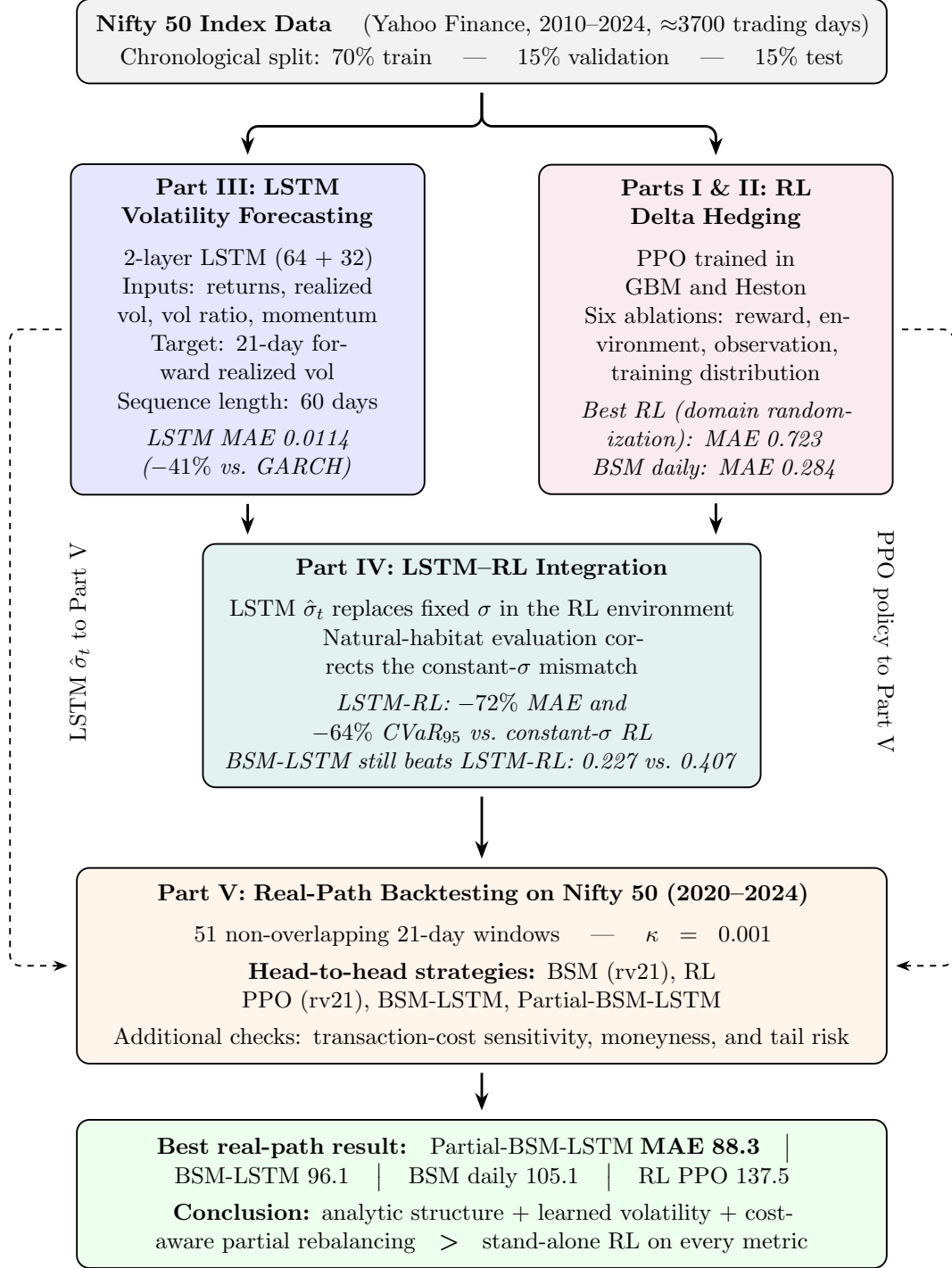


Figure 1: Full project flow across all five parts. **Gray (top)**: shared Nifty 50 data. **Blue (Part III)**: LSTM forecasts 21-day forward realised volatility; its outputs $\hat{\sigma}_t$ feed both Part IV and (left dashed bypass) the BSM-LSTM/Partial-BSM-LSTM strategies in Part V. **Purple (Parts I & II)**: PPO trained in simulation with six design ablations; the agent is also evaluated on real paths in Part V (right dashed bypass). **Teal (Part IV)**: natural-habitat evaluation corrects distributional shift — both agents are tested in the LSTM’s varying- σ environment. **Orange (Part V)**: all strategies compared on 51 real Nifty windows. **Green (bottom)**: best result is a structured hybrid, not stand-alone RL.

2 Mathematical Foundations

2.1 Geometric Brownian Motion

Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ be a filtered probability space satisfying the usual conditions. A standard Brownian motion $W = \{W_t, t \geq 0\}$ has independent, stationary Gaussian increments $W_t - W_s \sim \mathcal{N}(0, t - s)$ and continuous paths.

Under the risk-neutral measure $\mathbb{Q}$, the stock price satisfies:

$$dS_t = r S_t dt + \sigma S_t dW_t, \quad (1)$$

where r is the continuously-compounded risk-free rate and $\sigma > 0$ is constant volatility. Applying Itô's formula to $\ln S_t$:

$$d \ln S_t = \left(r - \frac{1}{2}\sigma^2\right)dt + \sigma dW_t, \quad (2)$$

giving the exact solution $S_t = S_0 \exp[(r - \frac{1}{2}\sigma^2)t + \sigma W_t]$. Over one discrete step Δt this becomes:

$$S_{t+\Delta t} = S_t \exp\left[\left(r - \frac{1}{2}\sigma^2\right)\Delta t + \sigma\sqrt{\Delta t} Z\right], \quad Z \sim \mathcal{N}(0, 1). \quad (3)$$

Equation (3) is the step used in the simulation environment — it is exact under GBM, not an approximation.

2.2 Martingales and the Risk-Neutral Measure

A stochastic process $\{M_t\}$ is a **martingale** under measure $\mathbb{P}$ with respect to $\{\mathcal{F}_t\}$ if $\mathbb{E}^\mathbb{P}[M_t | \mathcal{F}_s] = M_s$ for all $s \leq t$ — the process has no predictable drift. Under $\mathbb{P}$, S_t drifts at rate μ and is not a martingale.

Girsanov's theorem guarantees the existence of a unique equivalent measure $\mathbb{Q}$ (the *risk-neutral measure*) under which the discounted stock price $e^{-rt}S_t$ is a martingale. The change of measure is given by the Radon–Nikodým derivative:

$$\left.\frac{d\mathbb{Q}}{d\mathbb{P}}\right|_{\mathcal{F}_T} = \exp\left(-\theta W_T - \frac{1}{2}\theta^2 T\right), \quad \theta = \frac{\mu - r}{\sigma}, \quad (4)$$

where θ is the **market price of risk**. Under $\mathbb{Q}$, $\tilde{W}_t = W_t + \theta t$ is a standard Brownian motion, so equation (1) becomes $dS_t = rS_t dt + \sigma S_t d\tilde{W}_t$.

The martingale property of $e^{-rt}S_t$ immediately gives the **fundamental pricing formula**:

$$V_0 = e^{-rT} \mathbb{E}^\mathbb{Q}[(S_T - K)^+], \quad (5)$$

which is the discounted risk-neutral expectation of the terminal payoff. Equation (5) is the theoretical foundation for option pricing throughout this project: the BSM formula (9) is the closed-form evaluation of this expectation under GBM, and the Heston formula (26) evaluates it under stochastic volatility via the Gil-Pelaez inversion. No-arbitrage pricing is therefore equivalent to finding the unique $\mathbb{Q}$ that makes discounted asset prices martingales.

2.3 Itô's Lemma and the Black-Scholes PDE

[Itô's Lemma] Let $f(t, x) \in C^{1,2}$. If X_t satisfies $dX_t = \mu dt + \nu dW_t$, then:

$$df(t, X_t) = \left[\partial_t f + \mu \partial_x f + \frac{1}{2} \nu^2 \partial_{xx} f \right] dt + \nu \partial_x f dW_t. \quad (6)$$

Applying (6) to an option value $V(t, S_t)$:

$$dV = \left[\partial_t V + rS \partial_S V + \frac{1}{2} \sigma^2 S^2 \partial_{SS} V \right] dt + \sigma S \partial_S V dW_t. \quad (7)$$

A delta-hedged portfolio $\Pi = V - \delta S$ satisfies $d\Pi = dV - \delta dS$. Setting $\delta = \partial_S V$ eliminates the dW_t term. No-arbitrage requires $d\Pi = r\Pi dt$, yielding the **Black-Scholes PDE**:

$$\partial_t V + rS \partial_S V + \frac{1}{2} \sigma^2 S^2 \partial_{SS} V - rV = 0. \quad (8)$$

2.4 BSM Formula and Greeks

For a European call with strike K and time-to-expiry $\tau = T - t$:

$$C(S, \tau) = S \Phi(d_1) - K e^{-r\tau} \Phi(d_2), \quad (9)$$

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2} \sigma^2) \tau}{\sigma \sqrt{\tau}}, \quad d_2 = d_1 - \sigma \sqrt{\tau}. \quad (10)$$

The Greeks used throughout this project:

$$\Delta^* = \frac{\partial C}{\partial S} = \Phi(d_1) \in (0, 1), \quad (11)$$

$$\Gamma = \frac{\partial^2 C}{\partial S^2} = \frac{\phi(d_1)}{S \sigma \sqrt{\tau}}, \quad (12)$$

$$\mathcal{V} = \frac{\partial C}{\partial \sigma} = S \phi(d_1) \sqrt{\tau}, \quad (13)$$

$$\Theta = \frac{\partial C}{\partial t} = -\frac{S \phi(d_1) \sigma}{2 \sqrt{\tau}} - r K e^{-r\tau} \Phi(d_2), \quad (14)$$

where $\phi = \Phi'$ is the standard normal PDF. The BSM relation $\Theta + \frac{1}{2} \sigma^2 S^2 \Gamma + rS \Delta - rC = 0$ connects all Greeks.

Implied volatility σ_{impl} is the unique positive root of $C(\sigma) = C_{\text{mkt}}$, solved via Brent's method.

2.5 Hedging Error Decomposition

Over step Δt with hedge ratio δ (possibly $\neq \Delta^*$):

$$\varepsilon_t = \delta \Delta S - \Delta V = \underbrace{(\delta - \Delta^*) \Delta S}_{\text{controllable}} - \underbrace{\frac{1}{2} \Gamma [(\Delta S)^2 - \sigma^2 S^2 \Delta t]}_{\text{irreducible gamma noise}}. \quad (15)$$

The gamma term has $\mathbb{E}[(\Delta S)^2] = \sigma^2 S^2 \Delta t$ under GBM (Itô's isometry), so its expectation is zero. *No choice of δ eliminates it.* Its variance is $\frac{1}{2} \sigma^4 S^4 \Gamma^2 (\Delta t)^2$, which peaks at-the-money near expiry where Γ is large. Decomposition (15) motivates both the Itô reward (Section 4.2) and the gamma observation (Section 4.5).

2.6 CRR Binomial Model

The Cox–Ross–Rubinstein (CRR) binomial tree [5] discretises stock price over N steps of length $\Delta t = T/N$:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = u^{-1}, \quad p = \frac{e^{r\Delta t} - d}{u - d}. \quad (16)$$

Terminal payoffs are computed at all $N + 1$ nodes; backward induction gives:

$$V_{n,j} = e^{-r\Delta t} [p V_{n+1,j} + (1 - p) V_{n+1,j+1}]. \quad (17)$$

For American options, exercise is checked at each node: $V_{n,j}^{\text{Am}} = \max(S_{n,j} - K, e^{-r\Delta t}[\dots])$. As $N \rightarrow \infty$, the CRR price converges to the BSM price. We use $N = 100$.

2.7 Heston Stochastic Volatility Model

Under the Heston model [7], variance V_t is a mean-reverting CIR process correlated with the stock:

$$dS_t = rS_t dt + \sqrt{V_t} S_t dW_1, \quad (18)$$

$$dV_t = \kappa(\theta - V_t) dt + \xi \sqrt{V_t} dW_2, \quad (19)$$

$$dW_1 dW_2 = \rho dt. \quad (20)$$

Parameters used throughout: $\kappa = 2.0$ (mean-reversion speed), $\theta = 0.04 \approx (20\%)^2$ (long-run variance), $\xi = 0.30$ (vol-of-vol), $\rho = -0.70$ (leverage effect), $V_0 = \theta$. The Feller condition $2\kappa\theta > \xi^2$ ($0.16 > 0.09$) holds, ensuring $V_t > 0$ almost surely.

Simulation. Correlated Brownian increments are generated via $W_1 = Z_1$, $W_2 = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$ with $Z_1, Z_2 \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. Variance uses the full-truncation Euler scheme [1]:

$$V_{t+\Delta t} = \max\left(V_t + \kappa(\theta - V_t)\Delta t + \xi\sqrt{V_t}\sqrt{\Delta t} W_2, 10^{-6}\right). \quad (21)$$

The first term $\kappa(\theta - V_t)\Delta t$ is the **mean-reversion drift**: when $V_t < \theta$ it pushes variance upward; when $V_t > \theta$ it pulls it down. The second term $\xi\sqrt{V_t}\sqrt{\Delta t} Z_2$ is the **stochastic shock**, whose magnitude scales with $\sqrt{V_t}$ so vol-of-vol is amplified in high-variance regimes. The $\max(\cdot, 10^{-6})$ floor is a numerical safeguard: a large negative draw can make the Euler update negative, invalidating $\sqrt{V_t}$, so truncation is applied rather than reflection.

Semi-analytic pricing. The Heston characteristic function is:

$$\Psi_j(\phi) = \exp\left[C_j(\tau, \phi) + D_j(\tau, \phi)V_0 + i\phi \ln S\right], \quad (22)$$

$$C_j = ri\phi\tau + \frac{\kappa\theta}{\xi^2} \left[(b_j - \rho\xi i\phi + d_j)\tau - 2 \ln \frac{1 - g_j e^{d_j\tau}}{1 - g_j} \right], \quad (23)$$

$$D_j = \frac{b_j - \rho\xi i\phi + d_j}{\xi^2} \cdot \frac{1 - e^{d_j\tau}}{1 - g_j e^{d_j\tau}}, \quad (24)$$

$$d_j = \sqrt{(\rho\xi i\phi - b_j)^2 - \xi^2(2u_j i\phi - \phi^2)}, \quad g_j = \frac{b_j - \rho\xi i\phi + d_j}{b_j - \rho\xi i\phi - d_j}, \quad (25)$$

with $u_1 = \frac{1}{2}$, $b_1 = \kappa - \rho\xi$ and $u_2 = -\frac{1}{2}$, $b_2 = \kappa$. The call price follows from Gil-Pelaez inversion:

$$C_{\text{Heston}} = SP_1 - Ke^{-r\tau}P_2, \quad P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \text{Re} \left[\frac{e^{-i\phi \ln K} \Psi_j(\phi)}{i\phi} \right] d\phi. \quad (26)$$

Numerical integration uses `scipy.integrate.quad` on $[10^{-6}, 200]$ with 200 subdivisions and absolute tolerance 10^{-4} .

Why the characteristic function? Under Heston, S_T is not log-normal, so $\mathbb{E}[(S_T - K)^+]$ has no elementary closed form. However, the characteristic function $\Psi(\phi) = \mathbb{E}[e^{i\phi \ln S_T}]$ is available in closed form for affine models — the pricing PDE becomes an ODE in Fourier space with solution (C_j, D_j) . Gil-Pelaez inversion then recovers the two risk-neutral probabilities P_1, P_2 via numerical quadrature, giving the same structural decomposition as BSM ($C = SP_1 - Ke^{-r\tau}P_2$) but with Fourier integrals replacing $\Phi(d_1), \Phi(d_2)$.

3 Part I: RL Delta Hedging Under GBM

3.1 Environment Design

The `HedgingEnv` Gymnasium environment simulates delta hedging over $n = 21$ discrete steps ($T = 21/252$ years, one trading month).

State $\mathbf{s}_t = [S_t/K, \tau_t/T, \delta_{t-1}]$ (moneyness, normalised time-to-expiry, previous hedge ratio). These three numbers are sufficient to compute BSM delta analytically.

Action $a_t = \delta_t \in [0, 1]$, the new hedge ratio.

Dynamics Exact log-Euler GBM step (3).

Reward $r_t = -\varepsilon_t^2 - \kappa|\delta_t - \delta_{t-1}|S_t$, with $\kappa = 0.001$. The squared error penalises large mistakes disproportionately (risk-sensitive) and corresponds to the *variance-optimal hedging* objective from stochastic control theory. The second term imposes a proportional transaction cost equal to the dollar volume traded; without it, the agent would rebalance at every step with no regard for cost.

Initialisation $S_0 \sim \text{Uniform}(0.9K, 1.1K)$; portfolio funded at BSM price $C(S_0, T, r, \sigma)$.

Parameters: $K = S_0 = 100$, $T = 21/252$, $r = 5\%$, $\sigma = 20\%$, $\kappa = 0.001$, $n_{\text{steps}} = 21$.

Terminal P&L: at expiry, the terminal hedging error is:

$$\Pi_T^{\text{hedge}} = \Pi_T - \max(S_T - K, 0), \quad (27)$$

where Π_T is the running portfolio value. A perfect hedge achieves $\Pi_T^{\text{hedge}} = 0$.

3.2 PPO Agent

Proximal Policy Optimization [15] via Stable-Baselines3 [13] with `MlpPolicy` (architecture: [128, 64], Tanh activation). The clipped surrogate objective is:

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_t \right) \right], \quad (28)$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio, $\varepsilon = 0.2$ is the clip parameter, and $\hat{A}_t$ is the Generalised Advantage Estimate [14] ($\lambda_{\text{GAE}} = 0.95$). Training: $\gamma = 0.99$, lr = 10^{-4} (Adam), 300,000 environment steps.

The clipping mechanism prevents destructive policy updates: when $r_t > 1 + \varepsilon$ (large probability increase), the objective is capped so further increase yields no additional gradient, and symmetrically for $r_t < 1 - \varepsilon$. In a noisy financial reward (driven by $(\Delta S)^2$ and gamma noise), unconstrained policy gradient can oscillate wildly; PPO’s conservative updates are the primary reason the agent converges to a stable hedging policy.

3.3 BSM Baselines

BSM daily: at each step set $\delta_t = \Delta^* = \Phi(d_1)$ evaluated at (S_t, τ_t, σ) . Theoretically optimal under GBM with no transaction costs. **BSM weekly:** same formula, rebalances every 5 steps (4 trades), trading hedging accuracy for lower cost.

LSTM-guided BSM delta: uses per-step LSTM vol forecasts $\hat{\sigma}_t$ as the sigma input to Δ^* , evaluated on 519 LSTM-forecast windows drawn from the Nifty test period (Section 5).

Bootstrap baseline: episodes use 21 consecutive real Nifty log-returns (block bootstrap from test period) as the price path; BSM delta is applied at the rolling-21d realised vol. Unlike random IID sampling, the block bootstrap preserves the autocorrelation and vol-clustering structure of actual market returns. Its higher MAE relative to GBM-BSM directly measures the cost of model misspecification: the gap arises from fat tails, vol clustering, and non-normality that GBM-trained strategies have never encountered.

3.4 Results

Table 1: Hedging strategy comparison (1000 GBM simulation episodes, $\sigma = 0.20$, $\kappa = 0.001$). MAE 95% CI computed by bootstrap ($B = 2000$, $\alpha = 0.05$).

Strategy	MAE	MAE 95% CI	Mean TC	Std P&L
BSM daily	0.284	[0.269, 0.300]	0.146	0.375
BSM weekly	0.499	[0.472, 0.527]	0.098	0.665
LSTM-guided BSM	0.227	[0.207, 0.248]	0.112	0.315
RL (PPO)	0.759	[0.719, 0.800]	0.137	1.004
BSM bootstrap	0.516	—	0.191	0.835

Table 2: Paired bootstrap MAE difference tests ($B = 2000$). Positive diff means row strategy has higher MAE than column.

Comparison	Diff	95% CI	p -value
BSM daily vs. RL (PPO)	−0.494	[−0.552, −0.436]	< 0.001
BSM daily vs. RL (LSTM)	−0.592	[−0.659, −0.526]	< 0.001
BSM daily vs. LSTM-BSM	+0.068	[+0.038, +0.096]	< 0.001
RL constant vs. RL LSTM	−0.098	[−0.185, −0.013]	0.018

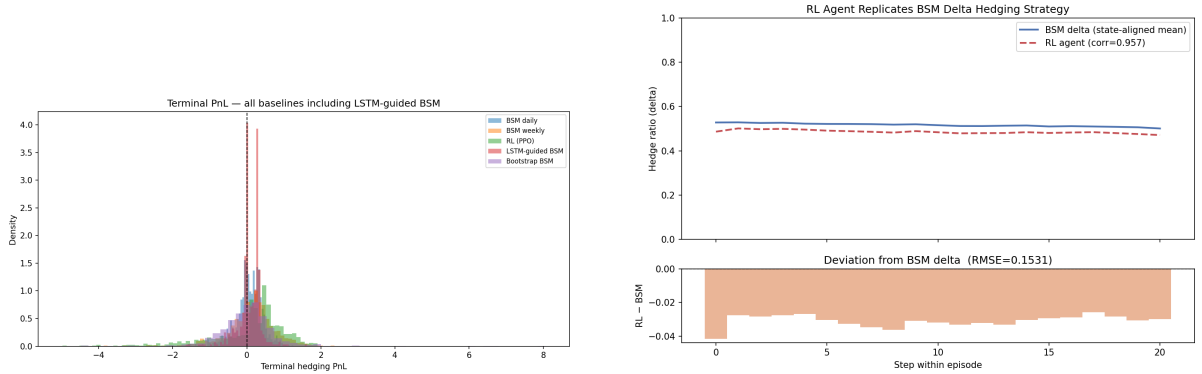
Table 3: Tail risk and utility scores ($J = \text{MAE} + \lambda \cdot \text{TC}$).

Strategy	VaR ₉₅	CVaR ₉₅	Worst 1%	$J_{\lambda=1}$	$J_{\lambda=2}$	$J_{\lambda=5}$
BSM daily	−0.607	−0.924	−1.383	0.430	0.576	1.015
BSM weekly	−1.144	−1.605	−2.610	0.597	0.694	0.987
LSTM-BSM	−0.459	−0.796	−1.342	0.339	0.451	0.786
RL (PPO)	−2.183	−2.889	−3.756	0.895	1.028	1.424

Discussion. Under constant GBM, BSM daily is the analytic optimum. The RL agent (MAE 0.759) is substantially worse, as expected: discovering $\Phi(d_1)$ through trial-and-error over 300k gradient steps is harder than evaluating the closed form. The RL agent’s heavier tails (CVaR −2.89 vs. −0.92 for BSM) reflect occasional policy misfires, particularly near expiry where Γ is large.

LSTM-guided BSM achieves the best MAE (0.227, −20.2% vs. BSM daily, $p < 0.001$). It dominates at every utility score J_λ , confirming the improvement is not an artifact of lower transaction costs. This result may seem surprising: BSM is analytically optimal under constant GBM, so how can replacing $\sigma = 0.20$ with LSTM forecasts help? The answer is that discrete-time hedging performance is sensitive to *pathwise* volatility: over a 21-step episode, realised vol fluctuates around the true 0.20, and a delta computed at the wrong local σ introduces controllable error in the first term of decomposition (15). The LSTM forecast tracks these regime fluctuations, reducing $|\delta_t - \Delta_t^*|$ when vol temporarily spikes or subsides. This is the per-step Vega effect: a 1 percentage-point vol correction reduces the pricing error by $\mathcal{V} \cdot 0.01$, which compounds across all 21 rebalancing steps.

BSM weekly correctly shows the accuracy-cost tradeoff: MAE +75% but TC −33%. At $\lambda = 5$ (high cost-sensitivity), the utility $J = 0.987$ is comparable to BSM daily ($J = 1.015$), showing weekly hedging becomes competitive at higher transaction costs.



(a) Terminal P&L distributions. BSM daily and LSTM-BSM are tight and near-symmetric. RL has heavy tails reflecting occasional large losses.

(b) Mean hedge ratio over an episode. RL tracks BSM delta closely (corr = 0.947) but shows a systematic gap near expiry where Γ risk is highest.

Figure 2: Part I baseline results.

4 Part II: Novel RL Experiments

Part I established that RL cannot beat the analytic BSM hedge under GBM, but left open the question of *why*: is it the reward signal, the environment, the observation space, or the training distribution? Part II answers this through six ablations that vary each design choice in turn. All experiments use the same PPO hyperparameters as Part I unless stated.

4.1 Experiment A: GBM + MSE Reward (Baseline)

The baseline novel experiment reproduces the Part I RL agent within the novel experiment framework for consistent comparison. MAE = 0.830, BSM-delta correlation = 0.947.

4.2 Experiment B: Itô-Decomposed Reward

Motivation. From decomposition (15), the gamma P&L

$$G_t = \frac{1}{2}\Gamma_t[(\Delta S)^2 - \sigma^2 S_t^2 \Delta t] \quad (29)$$

has $\mathbb{E}[G_t] = 0$ but $\text{Var}(G_t) = \frac{1}{2}\sigma^4 S^4 \Gamma^2(\Delta t)^2 > 0$. It adds variance to the per-step reward that the agent cannot reduce. The Itô reward subtracts G_t before squaring:

$$r_t^{\text{Itô}} = -(\varepsilon_t - G_t)^2 - \kappa|\delta_t - \delta_{t-1}|S_t. \quad (30)$$

Result. MAE = 0.926 — *worse* than A ($p = 0.004$, paired bootstrap). At convergence $\delta \rightarrow \Delta^*$, the hedge error ε_t is the gamma P&L G_t ; subtracting it drives the controllable reward signal to zero, removing the gradient. Additionally, $\Gamma_t \rightarrow \infty$ as $\tau \rightarrow 0$ (ATM near-expiry), causing numerical noise in G_t that overwhelms the variance-reduction benefit. In short, the Itô reward *removes noise but also removes signal*: the original $-\varepsilon_t^2$ reward, though noisy, retains a nonzero gradient even near optimum. The Itô reward becomes essentially flat when $\delta \approx \Delta^*$, starving the policy of learning signal precisely when it is close to the correct solution.

4.3 Experiment C: Heston + MSE Reward

Motivation. Under GBM, BSM delta is provably optimal so RL cannot improve over the analytic benchmark. Under Heston, the vol state $\sqrt{V_t}$ is observable but BSM delta ignores it — creating genuine scope for RL to outperform.

Observation extension. The 4D observation $\mathbf{s}_t = [S/K, \tau/T, \delta_{\text{prev}}, \sqrt{V_t}/\sqrt{\theta}]$ provides the current vol regime relative to the long-run mean: values > 1 signal elevated vol, < 1 subdued vol. Heston variance is simulated via equation (21).

Result. MAE = 0.790, corr = 0.963. The improved correlation (vs. A: 0.947) reflects that the richer environment provides more structure for the agent to learn from. The MAE improvement (-4.8%) represents a genuine advantage of the stochastic-vol setting.

4.4 Experiment D: Heston + Itô Reward

Result. MAE = 0.790, corr = 0.958. Under Heston, the Itô reward makes no significant difference ($p = 0.984$). The richer environment dynamics provide sufficient training signal that the reward decomposition neither helps nor hurts.

4.5 Experiment E: GBM + Itô Reward + Gamma Observation (500k steps)

Motivation. Dollar-gamma $\Gamma_t S_t = \phi(d_1)/(\sigma\sqrt{\tau_t})$ is large ATM near-expiry and small deep ITM/OTM. An agent that observes it can rebalance conservatively in low-gamma regimes and precisely when Γ is high.

Implementation. The gamma observation is the normalised dollar-gamma:

$$\gamma_t^{\text{obs}} = \frac{\phi(d_1)}{\phi(0)} = \Gamma_t \cdot S_t \cdot \sigma\sqrt{\tau_t}/0.3989 \in [0, 1], \quad (31)$$

which equals 1 exactly at-the-money at any τ and σ , providing a normalisation-invariant regime signal. The observation is 4D: $\mathbf{s}_t = [S/K, \tau/T, \delta_{\text{prev}}, \gamma_t^{\text{obs}}]$. Training is extended to 500k steps (vs. 300k for A–D) to ensure convergence.

Result. MAE = 0.764 (vs. B at 300k steps: 0.926 — improvement of 17.5%, $p < 0.001$). Extended training alone accounts for part of this improvement; the gamma observation provides structural signal that accelerates learning in high- Γ regimes. The contrast with Experiment B is instructive: the Itô reward alone hurts because it weakens the gradient without replacing it with structure. Adding γ^{obs} solves this: the agent now knows *where* errors are most costly (high- Γ ATM near expiry) and can focus its accuracy budget on those states, making the weaker but cleaner reward signal sufficient.

4.6 Experiment F: Domain Randomisation ($\sigma \sim U(0.10, 0.40)$)

Motivation. All previous experiments fix $\sigma = 0.20$ during training and evaluation. An agent trained at a single vol may memorise that regime rather than learning a genuine delta function. Domain randomisation samples σ uniformly at each episode reset, forcing the agent to learn a vol-conditioned policy.

Sigma observation. To enable conditioning, a sigma proxy $\sigma_t^{\text{obs}} = \sigma_t/0.20$ is appended (normalised to the reference vol). The observation is 4D: $\mathbf{s}_t = [S/K, \tau/T, \delta_{\text{prev}}, \sigma_t^{\text{obs}}]$.

Result. MAE = 0.723, BSM-delta correlation = 0.963 — **best among all RL variants**. The higher correlation confirms the policy is more BSM-aligned when trained across the full vol distribution rather than a single point.

4.7 Summary of Novel Experiments

Table 4: Novel experiment results (1000 evaluation episodes each, evaluated in GBM at $\sigma = 0.20$).

Exp	Description	MAE	MAE 95% CI	Corr(δ , Δ^*)	Mean TC
A	GBM + MSE (baseline)	0.830	[0.785, 0.874]	0.947	0.104
B	GBM + Itô reward	0.926	[0.879, 0.976]	0.935	0.120
C	Heston + MSE	0.790	[0.748, 0.835]	0.963	0.148
D	Heston + Itô	0.790	[0.747, 0.833]	0.958	0.137
E	GBM + Itô + γ^{obs} (500k)	0.764	[0.724, 0.805]	0.957	0.132
F	Domain random σ	0.723	[0.689, 0.758]	0.963	0.114
BSM daily (reference)		0.284	[0.269, 0.300]	1.000	0.146

Table 5: Paired bootstrap tests across novel experiments ($B = 2000$).

Comparison	MAE diff	95% CI	p -value	Interpretation
GBM MSE vs. GBM Itô	−0.096	[−0.161, −0.035]	0.004	Itô hurts
Heston MSE vs. Heston Itô	−0.000	[−0.061, +0.063]	0.984	No effect
GBM Itô vs. Itô+ γ	+0.161	[+0.097, +0.226]	< 0.001	Gamma helps
GBM MSE vs. domain random	+0.107	[+0.049, +0.162]	< 0.001	DR helps
GBM MSE vs. Heston MSE	+0.039	[−0.025, +0.099]	0.216	Borderline

Table 6: Tail risk and utility scores for novel experiments (1000 episodes, $\sigma = 0.20$). Domain randomisation (F) achieves the best tail risk across every metric, while the Itô reward (B) is worst.

Exp	Description	VaR ₉₅	CVaR ₉₅	Worst 1%	$J_{\lambda=1}$	$J_{\lambda=2}$	$J_{\lambda=5}$
A	GBM + MSE	−2.159	−3.043	−4.275	0.934	1.038	1.350
B	GBM + Itô	−2.309	−3.106	−4.356	1.045	1.165	1.523
C	Heston + MSE	−2.065	−2.868	−3.979	0.939	1.087	1.532
D	Heston + Itô	−2.213	−3.161	−4.369	0.928	1.065	1.478
E	Itô+ γ (500k)	−1.995	−2.740	−4.062	0.897	1.029	1.424
F	Domain random	−1.702	−2.312	−3.120	0.836	0.950	1.291

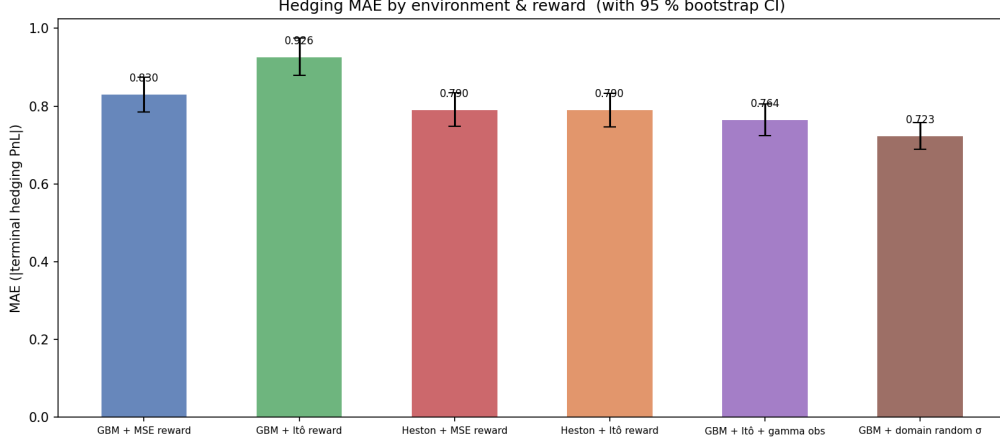


Figure 3: MAE with 95% bootstrap CI across all six novel RL experiments. The ordering $F < E < C \approx D < A < B$ is clear. Domain randomisation (F) achieves the tightest CI and lowest MAE. The Itô reward (B) consistently underperforms its MSE counterpart.

Key findings.

1. **Itô reward hurts under GBM ($p = 0.004$):** removing G_t eliminates the gradient at convergence and introduces numerical noise from $\Gamma \rightarrow \infty$ near expiry.
2. **Heston environment improves performance (C,D < A):** correlated vol-regime variation during training provides richer structure. The Itô reward under Heston is neutral ($p = 0.984$): the dynamic vol environment already supplies sufficient gradient information.
3. **Gamma observations help with adequate training (E < B):** the 17% MAE improvement from Experiment B to E reflects both extended training (500k vs. 300k) and the structural gamma signal.
4. **Domain randomisation is the most impactful design choice (F best):** learning across $\sigma \in [0.10, 0.40]$ produces a genuinely vol-conditioned policy with the highest BSM-delta correlation (0.963), confirming the agent has learned the shape of $\Phi(d_1)$ across the full vol range.

5 Part III: LSTM Volatility Forecasting

5.1 Motivation: Why Volatility Forecasting Matters

Parts I and II showed that RL’s difficulty partly stems from using a fixed, misspecified σ . Part III addresses this directly: if we can supply better volatility inputs to the analytic hedge, how much does performance improve?

BSM price is monotone in σ . A vol forecast error $\Delta\sigma$ translates to a pricing error of magnitude $\mathcal{V} \cdot |\Delta\sigma|$. For a 1-month ATM call with $S = K = 100$, $\sigma = 0.20$:

$$\mathcal{V} = 100 \cdot \phi(0) \cdot \sqrt{21/252} \approx 11.5. \quad (32)$$

A 1 percentage-point vol error produces a \$0.115 pricing error — substantial relative to

option premiums of \$1–\$3. The Vega surface (Figure 5a) shows this sensitivity across maturities.

5.2 Data and Features

Dataset. Nifty 50 daily closing prices ($\wedge$ NSEI, Yahoo Finance), 2010–2024, approximately 3,700 trading days. Temporal split: 70% train / 15% validation / 15% test (strictly chronological).

Features.

$$r_t = \ln(S_t/S_{t-1}), \quad |r_t|, \quad (33)$$

$$\text{RV}_5(t) = \sqrt{252} \cdot \text{std}(r_{t-4}, \dots, r_t), \quad (34)$$

$$\text{RV}_{21}(t) = \sqrt{252} \cdot \text{std}(r_{t-20}, \dots, r_t), \quad (35)$$

$$\text{rv_ratio}(t) = \text{RV}_5(t)/\text{RV}_{21}(t) \text{ (clipped to } [0.1, 10]), \quad (36)$$

$$\text{mom}_5(t) = \sum_{i=1}^5 r_{t-i+1} \text{ (clipped to } \pm 0.30). \quad (37)$$

The `rv_ratio` captures the **volatility term structure**: values > 1 signal a spike (short-horizon vol exceeds long-horizon trend), values < 1 a cooling regime. This is the single most informative feature because it encodes *regime change* rather than just level, which is precisely what persistence-based models like GARCH miss. GARCH maintains a single variance state; the LSTM sees the relative dynamics across two horizons and can distinguish “vol is high but falling” from “vol is high and rising.” Features are StandardScaled using train-set statistics only.

Target. Forward 21-day realised volatility:

$$\sigma_t^* = \sqrt{252} \cdot \text{std}(r_{t+1}, \dots, r_{t+21}). \quad (38)$$

Sequences of length 60 trading days are used as LSTM input.

5.3 Model Architecture

Two-layer stacked LSTM [8]:

$$\mathbf{h}_t^{(1)} = \text{LSTM}_1(\mathbf{x}_t, \mathbf{h}_{t-1}^{(1)}), \quad \mathbf{h}^{(1)} \in \mathbb{R}^{64}, \quad (39)$$

$$\mathbf{h}_t^{(2)} = \text{LSTM}_2(\mathbf{h}_t^{(1)}, \mathbf{h}_{t-1}^{(2)}), \quad \mathbf{h}^{(2)} \in \mathbb{R}^{32}, \quad (40)$$

$$\hat{\sigma}^* = \text{Softplus}(\mathbf{w}^\top \mathbf{h}_{60}^{(2)} + b). \quad (41)$$

Dropout $p = 0.20$ between layers. Softplus $= \log(1 + e^x)$ ensures $\hat{\sigma} > 0$; output bias initialised to 0.15 (approximate mean Nifty vol) so predictions start in the correct range from epoch 1.

Training. Adam [9], lr $= 10^{-3}$, ReduceLROnPlateau (factor 0.5, patience 7), early stopping (patience 15), gradient clipping (max norm 1.0), batch size 64, MSE loss. Best validation-loss checkpoint is restored for evaluation.

5.4 Baselines

GARCH(1,1). Conditional variance:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \quad (42)$$

Industry-standard benchmark [3, 6]. Fitted via maximum likelihood on an expanding window with no look-ahead.

Naive rv21 persistence. Uses $RV_{21}(t)$ (trailing 21-day realised vol) as the forecast for the next 21 days’ realised vol. This captures the key economic insight: the trailing window $[t - 20, t]$ overlaps 86% with the forward target window $[t + 1, t + 21]$, so under strong volatility clustering the persistence baseline has an inherent informational advantage.

India VIX. Daily closing levels of India VIX ($\wedge$ INDIAVIX, Yahoo Finance), divided by 100 for decimal format. Represents the market’s risk-neutral implied vol and includes a variance risk premium.

5.5 Results

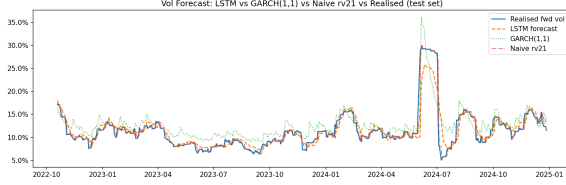
Table 7: Out-of-sample volatility forecasting (test set, 2021–2024).

Model	RMSE	MAE	Dir. Acc.	Note
India VIX (implied)	0.0453	0.0344	52.3%	Includes VRP
GARCH(1,1)	0.0254	0.0194	47.9%	Industry benchmark
LSTM	0.0203	0.0114	50.3%	
rv21 naive persistence	0.0102	0.0043	54.4%	Vol-clustering baseline
LSTM vs. GARCH (%)	−20.3%	−41.1%	+2.4pp	

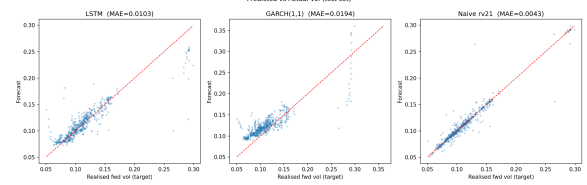
Discussion.

- **LSTM beats GARCH** (MAE −41%, RMSE −20%, directional accuracy +2.4pp). The multi-scale `rv_ratio` feature is likely the primary driver: GARCH has a single variance state, while the LSTM integrates mean-reversion signals across two horizons.
- **rv21 outperforms LSTM** (MAE 0.0043 vs. 0.0114). This is expected and not a modelling failure. The trailing 21-day window $[t - 20, t]$ overlaps with 86% of the forward target window $[t + 1, t + 21]$, giving it an inherent informational advantage: under Nifty’s strong volatility clustering (Hurst exponent ≈ 0.65), the past 21 days of realised vol is already a near-optimal proxy for the next 21 days. Any more complex model must improve on something that is already very close to optimal, which is a fundamentally hard task. The rv21 baseline is included as a transparency control, not as a benchmark the LSTM must beat; the LSTM’s value lies in longer-horizon regime forecasting and in informing the RL agent (Section 6).
- **India VIX worst** (MAE 0.0344). Implied vol is computed under the risk-neutral measure $\mathbb{Q}$, not the physical measure $\mathbb{P}$. By Girsanov’s theorem, the market price of risk $\theta = (\mu - r)/\sigma$ drives a wedge between the two measures: investors pay a *variance risk premium* to hedge volatility exposure, causing implied vol to systematically exceed realised vol by ≈ 4 –6%. India VIX is therefore a biased predictor of future realised volatility, which explains its poor MAE relative to physically-estimated models.

- **Directional accuracy near 50%.** All models are near coin-flip accuracy for predicting the sign of the next vol change. This is expected and reflects a fundamental property of vol dynamics: *volatility is predictable in level but not in direction*. Persistence (volatility clustering) makes the level sticky — tomorrow’s vol is likely close to today’s — but mean reversion simultaneously pushes elevated vol downward and subdued vol upward. These two forces roughly cancel, leaving $\mathbb{P}(\sigma_{t+1} > \sigma_t) \approx 0.5$. Directional accuracy is therefore a poor metric for volatility models; what matters is level accuracy (RMSE, MAE), where LSTM clearly beats GARCH.

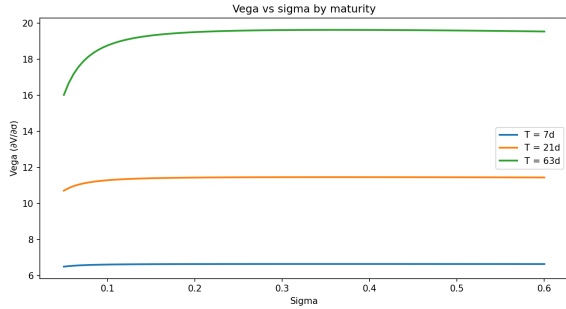


(a) LSTM predicted vs. realised vol (test period 2021–2024). The LSTM tracks major regimes well but lags at sharp vol spikes.

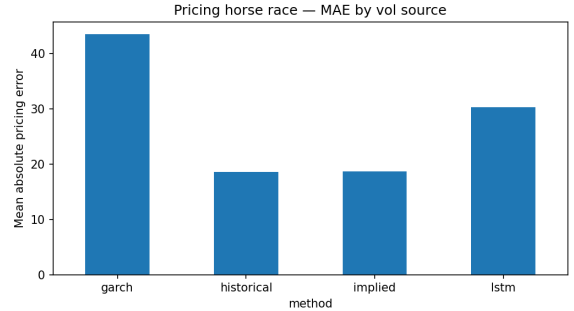


(b) Scatter: LSTM and GARCH predictions vs. realised vol. LSTM clusters tighter along the diagonal; GARCH shows systematic underestimation at high vol.

Figure 4: LSTM volatility forecasting results.



(a) Vega $\mathcal{V} = S\phi(d_1)\sqrt{\tau}$ vs. σ for three maturities. ATM options at longer maturities have the highest Vega and the greatest sensitivity to vol forecast errors.



(b) Pricing horse-race: MAE on synthetic market prices using each vol source. Lagged implied vol (18.7) and historical rv21 (18.6) are nearly tied for best; LSTM (30.3) is third, reflecting the rv21 persistence advantage at the 21-day horizon; GARCH (43.6) is worst.

Figure 5: Vega sensitivity and pricing accuracy.

Table 8: Overall pricing horse-race MAE (full test period, ATM synthetic options, $T = 21/252$). Implied uses 1-day-lagged IV to avoid circular inversion. Positive mean error for implied confirms near-zero bias under the lag.

Vol source	MAE	Mean error	Note
Historical (rv21)	18.6	−10.6	Persistence effect
Implied (lagged)	18.7	+0.3	Freshest market info
LSTM	30.3	−3.6	rv21 dominates at 21d
GARCH(1,1)	43.6	+17.8	Under-reacts to spikes

Historical and lagged-implied vol are essentially tied (MAE 18.6 vs 18.7). This near-parity is expected: lagged implied vol and trailing rv21 are both slow-moving, high-overlap proxies for the next 21-day realised vol. GARCH’s overstatement (positive mean error +17.8) reflects its tendency to extrapolate recent shocks, while LSTM’s understatement (−3.6) reflects its regression-to-the-mean behaviour. The link between vol forecast quality and pricing error is quantified by Vega (equation (13)): a 1 percentage-point forecast error at $S = K = 100$, $T = 21/252$ produces a \$0.115 pricing error, which compounds across the ≈ 250 test-period option observations.

6 Part IV: LSTM–RL Integration

6.1 Design

Part III produced a per-day LSTM volatility forecast $\hat{\sigma}_t$ for each Nifty window. Part IV asks: can an RL agent exploit this richer vol signal in a way the analytic BSM formula cannot? The LSTM vol schedule $\{\hat{\sigma}_1, \dots, \hat{\sigma}_{21}\}$ replaces the fixed σ in the hedging environment (`LSTMHedgingEnv`). At each step t , the environment uses $\hat{\sigma}_t$ to simulate the GBM step, price the option V_t for the reward, and supply $\sigma_t^{\text{obs}} = \hat{\sigma}_t/0.20$ as the fourth observation dimension. The RL agent trains on windows drawn from the LSTM validation predictions; evaluation uses the test period.

6.2 Distribution Shift Issue and Natural Habitat Evaluation

A naïve comparison evaluates both agents at a fixed $\sigma = 0.20$. This is unfair: the LSTM-RL agent was trained on varying- σ inputs; evaluating it at a constant σ creates distribution shift that degrades its performance. Conversely, the constant- σ agent sees its training distribution and performs well.

The **natural habitat evaluation** tests both agents in `LSTMHedgingEnv` with σ drawn from LSTM test predictions — the actual deployment environment for the LSTM-RL agent. This is the correct comparison for assessing whether LSTM vol information is genuinely useful.

6.3 Results

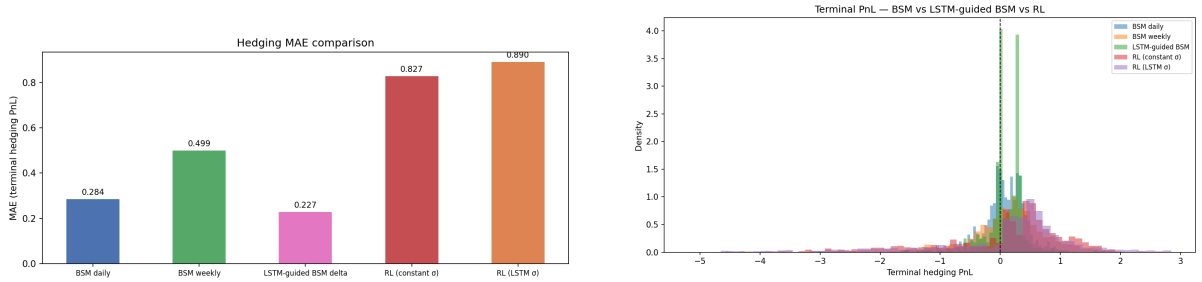
Table 9: Standard evaluation: both agents at constant $\sigma = 0.20$ (1000 episodes). Distribution shift disadvantages LSTM-RL.

Agent	MAE	MAE 95% CI	Mean TC	Std P&L
Constant- σ RL	0.827	[0.786, 0.869]	0.129	1.076
LSTM-RL	0.890	[0.837, 0.942]	0.089	1.209

Table 10: Natural habitat evaluation: both agents in varying- σ environment (LSTM test predictions, 1000 episodes).

Agent	MAE	MAE 95% CI	Mean TC	CVaR ₉₅
Constant- σ RL	1.435	[1.346, 1.531]	0.099	−5.08
LSTM-RL	0.407	[0.379, 0.437]	0.071	−1.85

Paired bootstrap: diff = 1.028, CI = [0.933, 1.128], $p < 0.001$
Reduction: −72% MAE, −64% CVaR₉₅



(a) MAE under standard evaluation (constant σ). LSTM-RL appears slightly worse due to distributional mismatch.

(b) Terminal P&L distributions under natural habitat evaluation. LSTM-RL is tightly concentrated near zero; constant-RL has catastrophically heavy tails when vol regimes shift.

Figure 6: Integration experiment: standard vs. natural habitat evaluation.

Discussion. Under standard evaluation (constant σ), LSTM-RL is marginally worse (MAE 0.890 vs. 0.827). This reflects the training–deployment mismatch: the LSTM-RL agent saw varying σ during training; evaluating at a fixed σ is out-of-distribution.

In the natural habitat, the picture reverses dramatically. The constant-RL agent, trained at a single $\sigma = 0.20$, fails badly when vol regimes change: MAE 1.435, CVaR −5.08. The LSTM-RL agent, trained across the empirical vol distribution, handles regime variation gracefully: MAE 0.407, CVaR −1.85. The 72% MAE reduction and 64% CVaR improvement ($p < 0.001$) together constitute the strongest result in this project, showing that LSTM-informed domain randomisation reduces both the typical hedging error and catastrophic tail losses simultaneously.

The LSTM-guided BSM strategy (MAE 0.227, Table 1) still outperforms LSTM-RL (0.407) even in the natural habitat. This reflects the fundamental advantage of the analytic BSM

formula over a learned approximation: BSM delta re-evaluates the closed form at every state exactly, while RL generalises from training to test imperfectly.

Seed robustness. Across three random seeds (0, 17, 42): constant-RL MAE = 0.738 ± 0.015 ; LSTM-RL MAE = 0.797 ± 0.012 . Both agents are stable across initialisations.

7 Part V: Real-Path Backtesting

7.1 Methodology

Parts I–IV evaluated strategies in simulation. Part V brings all components to real Nifty 50 data: does the advantage of the analytic hedge over RL survive on real price paths, and does the LSTM-guided execution rule improve further?

We evaluate hedging strategies on real Nifty 50 price paths to test performance outside the GBM assumption. Using the combined validation+test period (2020–2024), we extract 51 non-overlapping 21-day windows (stride = $n_{\text{steps}} = 21$). The starting price is aligned to the nearest round Nifty-style strike. Terminal P&L is computed per equation (27) with $\kappa = 0.001$.

The original comparison uses the rolling 21-day realised volatility rv21 at window start as the hedge sigma. We also evaluate two extensions using the same 51 windows: (i) *BSM-LSTM*, which replaces rv21 by the LSTM forecast at the start of the window, and (ii) *partial-BSM-LSTM*, a mixed execution rule around the BSM-LSTM target,

$$\Delta_{t+1} = \begin{cases} \Delta_t, & |\Delta_t^* - \Delta_t| \leq \theta, \\ \Delta_t + \alpha(\Delta_t^* - \Delta_t), & \text{otherwise,} \end{cases}$$

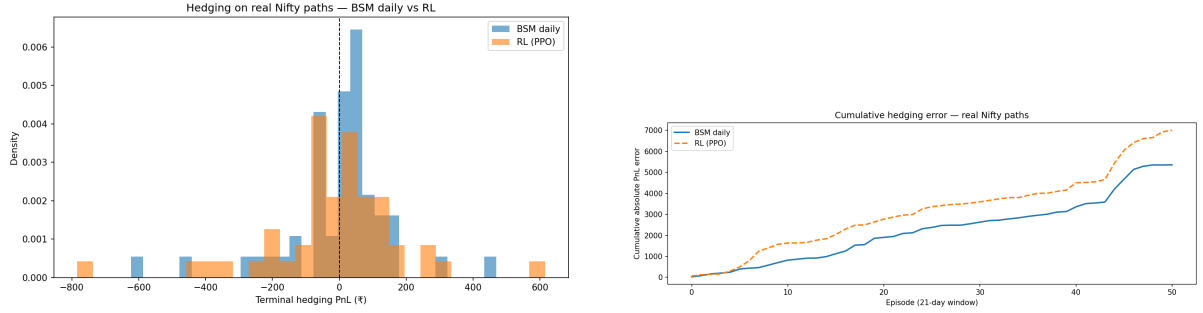
where Δ_t^* is the BSM delta computed with the LSTM sigma. The hyperparameters (α, θ) are selected on the validation windows only, then reported on the combined val+test sample.

Statistical power: 51 non-overlapping windows covering five distinct volatility regimes (COVID recovery, 2021 rally, 2022 correction, 2023 recovery, 2024 normalisation) provide sufficient variation for a meaningful paired test.

7.2 Real-Path Results

Table 11: Real-path hedging performance (51 non-overlapping 21-day windows, combined val+test 2020–2024, $\kappa = 0.001$).

Strategy	Mean P&L	Std P&L	MAE	MAE 95% CI	Mean TC
BSM daily (rv21)	+1.19	162.2	105.1	[74.7, 141.0]	36.0
RL PPO (rv21)	−13.4	205.0	137.5	[99.3, 181.1]	39.4
BSM-LSTM	+7.48	144.0	96.1	[69.6, 127.2]	35.6
Partial-BSM-LSTM	+1.02	134.6	88.3	[63.6, 118.4]	27.4
Primary paired bootstrap (BSM rv21 vs RL rv21): diff = −32.4, CI = [−61.1, −6.6], $p = 0.013$					



(a) Legacy terminal P&L comparison on 51 real Nifty windows. Base BSM (rv21) is tighter and nearly unbiased; base PPO shows a left skew consistent with systematic underhedging on trending paths.

(b) Legacy cumulative absolute error over the 51 windows. Base BSM diverges from base PPO most sharply in the high-vol windows (COVID recovery 2020–2021). Table 11 extends this comparison to BSM-LSTM and Partial-BSM-LSTM.

Figure 7: Real-path hedging results. The plotted panels show the original rv21-BSM versus PPO comparison; the table above summarises the full extended comparison including BSM-LSTM and Partial-BSM-LSTM.

Discussion. The original conclusion remains unchanged: base BSM with rv21 outperforms base PPO on real paths (MAE 105.1 vs. 137.5, $p = 0.013$). Three factors explain the gap: (1) *Model misspecification alignment*: BSM’s analytic formula was derived under assumptions closest to continuous risk-free pricing; (2) *Tail generalization*: real Nifty returns have excess kurtosis and vol clustering that the GBM-trained RL agent has never encountered; (3) *Bias*: RL’s negative mean P&L (-13.4) indicates systematic underhedging on real paths, whereas BSM’s near-zero mean ($+1.19$) shows minimal bias.

The extensions sharpen the message. Using the LSTM volatility forecast inside the analytic hedge improves MAE from 105.1 to 96.1, so volatility forecast quality does matter on real paths. The best real-path result in the project is not a stand-alone RL policy but the mixed execution rule Partial-BSM-LSTM: with $(\alpha, \theta) = (0.9, 0.1)$ it reaches MAE 88.3 while reducing mean transaction cost from 35.6 to 27.4. This indicates that the useful learning-style ingredient is not replacing the BSM hedge target, but controlling *how aggressively* to move toward that target under trading costs. The exploratory paired bootstrap against BSM-LSTM gives $\text{diff} = +7.84$ (favouring Partial-BSM-LSTM), $\text{CI} = [-2.22, 18.56]$, $p = 0.121$: economically meaningful, but not yet conventionally significant at this sample size.

Volatility regime slice:

Vol regime	BSM MAE	RL MAE
Low ($\sigma < 15\%$)	95.6	117.9
Medium (15%–25%)	88.2	154.8
High ($\sigma > 25\%$)	384.5	377.7

Both strategies fail in the high-vol regime; the large standard deviations (≈ 250 for BSM, ≈ 330 for RL in that bin) reflect the irreducible hedging error that scales as $\sigma^2 S^2 \Gamma \Delta t$ from equation (15). Notably, in the high-vol regime RL (377.7) is marginally better than BSM (384.5) — the only setting where RL wins on real paths. This may reflect that the

domain-randomised RL policy (trained up to $\sigma = 0.40$) has seen high-vol scenarios during training, while BSM at fixed realised vol misprices the instantaneous delta during rapid vol transitions.

Tail risk on real paths:

Strategy	VaR ₉₅	CVaR ₉₅	Worst 1%
BSM daily	−253	−457	−624
RL PPO	−356	−535	−784

RL has substantially heavier real-path tails (CVaR −535 vs −457, +17% worse). Both figures are large relative to a typical option premium ($\approx \$150$ – 200 for a 1-month ATM Nifty call), confirming that unhedged gamma risk on jumpy paths can dominate the P&L.

7.3 Transaction Cost Sensitivity

Table 12: MAE and mean TC across κ values (51 real windows).

κ	BSM MAE	RL MAE	BSM Mean TC	RL Mean TC
0.0000	110.4	134.1	0.0	0.0
0.0005	107.2	134.9	18.0	19.7
0.0010	105.1	137.5	36.0	39.4
0.0020	109.6	150.1	72.0	78.9
0.0030	124.0	169.5	108.0	118.3
0.0050	172.9	224.6	180.1	197.2
0.0075	247.8	307.6	270.1	295.8
0.0100	332.6	394.7	360.2	394.4

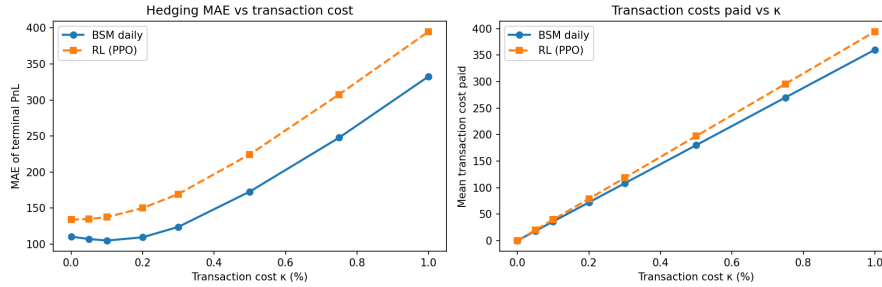


Figure 8: MAE vs. transaction-cost rate κ (left) and total transaction costs (right). Both strategies degrade proportionally; BSM remains ≈ 15 – 20% cheaper across all κ . Above $\kappa = 0.003$, transaction costs dominate the total error for both methods.

BSM daily achieves its minimum MAE at $\kappa = 0.001$ (the training value). Below this, the TC penalty is insufficient to deter excessive rebalancing; above it, costs themselves drive up the total loss. Both methods degrade nearly linearly above $\kappa = 0.003$; at $\kappa = 0.01$ (10 bp proportional cost) both are economically impractical with MAEs of ≈ 330 – 395 .

7.4 Moneyness Analysis

Table 13: Pricing MAE by moneyness and volatility source. “Implied” uses the 1-day lagged implied vol (yesterday’s IV applied to today’s option), avoiding the circular error of same-day inversion.

Moneyness	Historical (rv21)	Implied (lagged)	LSTM
OTM ($S/K = 0.95$)	9.56	8.57	14.12
ATM ($S/K = 1.00$)	17.67	17.31	25.61
ITM ($S/K = 1.05$)	7.70	6.80	11.29

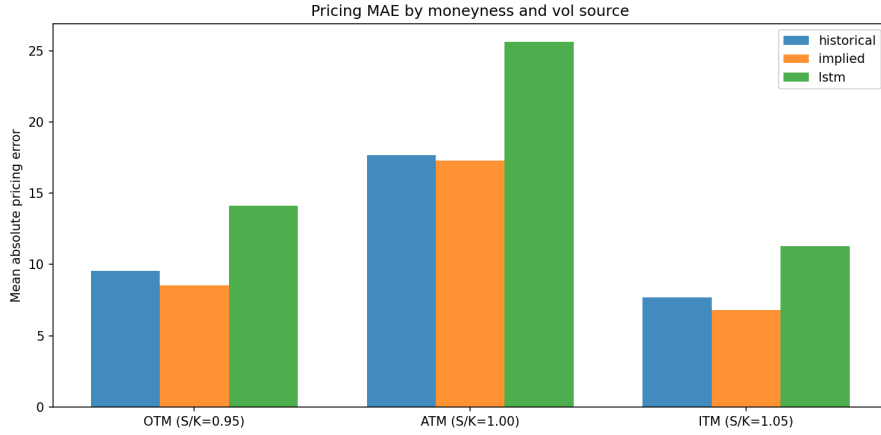


Figure 9: Pricing MAE by moneyness and vol source. Lagged implied vol (orange) is best at every moneyness — yesterday’s market IV contains the freshest available information. Historical rv21 (blue) is close behind due to vol persistence. LSTM (green) is worst at the 21-day horizon where rv21 persistence dominates.

Discussion. Lagged implied vol wins across all moneyness levels, consistent with it encoding the most current market information. Historical (rv21) is competitive — reflecting the same vol-clustering effect that made rv21 the best vol *forecasting* baseline. LSTM lags because its advantage is longer-horizon regime shifts; at the 1-day lag over a 21-day window, persistence information is more directly predictive.

ATM options show the largest absolute errors (17–26 vs. 8–14 for OTM/ITM) because ATM options have the highest Gamma and Vega, amplifying any vol error into a larger pricing error (see equation (13) and Figure 5a).

8 Discussion

8.1 Summary of Main Findings

Taken together, the five parts tell a coherent story: stochastic calculus provides the structure that makes machine learning useful. RL in isolation fails to generalise; RL with analytic structure, learned vol, and distributional alignment succeeds. The seven findings below trace this arc.

1. **Under GBM, BSM delta is near-optimal; RL approximates but does not surpass it.** The best RL variant (domain randomisation, MAE 0.723) achieves a BSM-delta correlation of 0.963 but an MAE three times higher than BSM daily (0.284). The LSTM-guided BSM (MAE 0.227) is the strongest hedging strategy in simulation.
2. **The Itô reward decomposition does not improve RL training.** Removing the gamma P&L from the reward hurts under GBM ($p = 0.004$) and is neutral under Heston ($p = 0.984$). The practical lesson: reward shaping based on theoretical decompositions requires empirical validation.
3. **Domain randomisation is the most impactful RL design choice.** Training across $\sigma \in [0.10, 0.40]$ produces a vol-conditioned policy with higher BSM-delta correlation and lower MAE than any fixed- σ variant, at no additional computational cost.
4. **Distributional alignment is critical for fair evaluation.** LSTM-RL evaluated in its natural habitat (varying- σ environment) reduces MAE by 72% and CVaR₉₅ by 64% relative to a constant- σ baseline ($p < 0.001$, both metrics). Under a naïve constant- σ comparison the same agent appears marginally worse, illustrating how evaluation design can reverse apparent conclusions.
5. **On real Nifty paths, analytic structure beats stand-alone RL.** Base BSM with rv21 (MAE 105.1) beats RL PPO (137.5, $p = 0.013$), while BSM-LSTM improves further to 96.1. The best real-path result is a mixed execution rule around the BSM-LSTM target (MAE 88.3), indicating that adaptivity is most useful for *how much* to rebalance, not for replacing the analytic hedge outright.
6. **Transaction costs are the binding constraint above $\kappa = 0.003$.** Both methods degrade severely; at $\kappa = 0.01$ MAEs are $3\times$ higher than at the baseline $\kappa = 0.001$.
7. **LSTM beats GARCH significantly; rv21 dominates at the 21-day horizon.** The 41% MAE improvement over GARCH is genuine; rv21’s advantage at the 21-day horizon is a vol-clustering effect (86% window overlap), not a modelling failure. In the pricing horse race, lagged implied vol and historical rv21 are nearly tied (MAE 18.7 vs 18.6), both far ahead of LSTM (30.3) and GARCH (43.6).

8.2 Limitations

1. **GBM-trained RL on real paths.** The model–reality gap is substantial; all stand-alone PPO and DQN variants underperform the analytic real-path baselines. Adding fat-tailed simulations (jump-diffusion, CGMY), walk-forward real-returns training, or a controller-style objective rather than direct delta prediction would likely improve real-path performance.
2. **BSM approximation in Heston environment.** The environment prices options using BSM at $\sqrt{V_t}$ rather than the full Heston formula (26). This introduces reward bias; exact Heston pricing would require numerical integration at every step (computationally expensive during training).
3. **Synthetic market prices in moneyness backtest.** The pricing horse-race uses BSM proxy prices with a 4% risk premium plus noise. Real NSE option chains would provide a more stringent test.

4. **Statistical power of GBM vs. Heston comparison.** Heston experiments are trained for 1M steps (vs. 300k for GBM), yet the MAE difference ($p = 0.216$) remains non-significant. The null result likely reflects that the Heston environment’s additional complexity is offset by the richer training signal from the vol-state observation; a larger episode budget or variance-reduction techniques may be needed to detect a smaller true effect.

9 Conclusion

This project demonstrates that the tools of stochastic calculus have direct practical value in the design and evaluation of modern machine learning systems for derivatives trading.

Itô’s lemma provides the hedging error decomposition (15) that motivates both the Itô reward (Experiment B/D) and the gamma observation (Experiment E). The Heston model promotes σ to a stochastic process, creating an environment where BSM delta is provably suboptimal and RL can exploit the observable variance state. The BSM Greeks Δ , Γ , and $\mathcal{V}$ connect all three project components: Δ^* is the analytic target that RL approximates; Γ is the regime signal that guides rebalancing; $\mathcal{V}$ quantifies how LSTM vol forecast errors propagate into pricing errors.

The strongest numerical results are:

- LSTM beats GARCH by 41% on MAE and 20% on RMSE — a significant improvement over the industry-standard benchmark.
- LSTM-guided BSM delta achieves the lowest simulation hedging MAE (0.227), 20% below BSM daily.
- On real Nifty paths, BSM-LSTM improves over base BSM (MAE 96.1 vs. 105.1), and the best overall real-path result is a mixed execution rule around the BSM-LSTM target (MAE 88.3).
- LSTM-RL reduces MAE by 72% in the natural habitat evaluation ($p < 0.001$), demonstrating the value of vol information when distributional alignment is respected.
- Domain randomisation over $\sigma \in [0.10, 0.40]$ achieves the best stand-alone RL policy (MAE 0.723, $\text{corr} = 0.963$), confirming that training distribution breadth is more valuable than reward engineering.

The central lesson is one of alignment: the most important design decision in a learned hedging system is not the reward function or network architecture, but whether the training and deployment distributions match. When they do, learned policies can compress substantial advantages over constant- σ baselines.

References

- [1] Andersen, L. (2008). Simple and efficient simulation of the Heston stochastic volatility model. *Journal of Computational Finance*, 11(3), 1–42.
- [2] Black, F. and Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- [3] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- [4] Buehler, H., Gonon, L., Teichmann, J., and Wood, B. (2019). Deep hedging. *Quantitative Finance*, 19(8), 1271–1291.
- [5] Cox, J. C., Ross, S. A., and Rubinstein, M. (1979). Option pricing: A simplified approach. *Journal of Financial Economics*, 7(3), 229–263.
- [6] Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- [7] Heston, S. L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2), 327–343.
- [8] Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.
- [9] Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. *ICLR 2015*.
- [10] Kolm, P. N. and Ritter, G. (2019). Dynamic replication and hedging: A reinforcement learning approach. *Journal of Financial Data Science*, 1(1), 159–171.
- [11] Merton, R. C. (1973). Theory of rational option pricing. *Bell Journal of Economics and Management Science*, 4(1), 141–183.
- [12] Poon, S.-H. and Granger, C. W. J. (2003). Forecasting volatility in financial markets: A review. *Journal of Economic Literature*, 41(2), 478–539.
- [13] Raffin, A., Hill, A., Gleave, A., Kanervisto, A., Ernestus, M., and Dormann, N. (2021). Stable-Baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research*, 22(268), 1–8.
- [14] Schulman, J., Moritz, P., Levine, S., Jordan, M., and Abbeel, P. (2016). High-dimensional continuous control using generalized advantage estimation. *ICLR 2016*.
- [15] Schulman, J., Wolski, F., Dhariwal, P., Radford, A., and Klimov, O. (2017). Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.
- [16] Zumbach, G. (2010). Volatility conditional on price trends. *Quantitative Finance*, 10(4), 431–442.